

Local affine-isometry charts

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Scope

Midpoint preservation yields preservation of affine segments on a smaller ball. The exact source path through omitted radial-map helpers constructs one ambient affine isometry agreeing with the given map there.

2 direct Cards + 5 reused prerequisites = 7 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Local affine-isometry chart on a smaller ball](#)
- Key result: [Affine-segment preservation on a quarter ball](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#)

Level 0 (2 Cards)

Level 0

Symmetric lens

Names the intersection of two closed balls with a common radius.

`MathlibAnnex.symmetricLens`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Center transport under bounded point-reflection symmetry

Uses a doubling argument to identify the reflection centers under a bijective isometry.

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (1 Card)

Level 1

Midpoint preservation on a symmetric lens

Identifies the centers of two lenses without prescribing images of their foci.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

Immediate Card prerequisites: [Center transport under bounded point-reflection symmetry](#) · [Symmetric lens](#)

Used by in this scope: [Midpoint preservation under lens containment](#)

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Level 2 (1 Card)

Level 2

Midpoint preservation under lens containment

Restricts a set isometry and its inverse to the two contained lenses.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Immediate Card prerequisites: [Midpoint preservation on a symmetric lens](#)

Used by in this scope: [Midpoint preservation on a quarter ball](#)

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Level 3 (1 Card)

Level 3

Midpoint preservation on a quarter ball

Supplies both lens inclusions from matching ambient balls.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation under lens containment](#)

Used by in this scope: [Affine-segment preservation on a quarter ball](#)

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Level 4 (1 Card)

Level 4

Affine-segment preservation on a quarter ball

Passes from midpoint preservation to every real parameter in a segment.

`MathlibAnnex.IsometryEquiv.map_lineMap_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation on a quarter ball](#)

Used by in this scope: [Local affine-isometry chart on a smaller ball](#)

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Level 5 (1 Card)

Level 5

Local affine-isometry chart on a smaller ball

Constructs one ambient surjective affine isometry from the same radial map throughout.

`MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

Immediate Card prerequisites: [Affine-segment preservation on a quarter ball](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)